

cosim-gen - SW IP Card

Schema Version		
Schema Version		0.9
Basic Info		
IP Name		cosim-gen
Version		0.9.0
Source Languages		C++, Python
TRL		TRL-6
Status		Prototype implemented
Provider	Organization Name	FZI Research Center for Information Technology
	Organization URL	www.fzi.de
	Organization Alias	FZI Forschungszentrum Informatik
License		Apache-2.0
License Details		With LLVM Exception
Repository URL		https://github.com/fzi-forschungszentrum-informatik/cosim-gen.git
Project		ISOLDE
Work Item		4.3
IP Type	Category	Software Tool, Domain Specific Simulation Tools, EDA Tools and methods for architecture exploration
	Domain	Other
	Comment	QEMU/RTL co-simulation generator: extracts individual peripherals out of a synthesizable CIRCT core dialects design into standalone SystemC models, wires each to QEMU over a remote-port channel, and runs QEMU alongside the RTL peripheral it talks to.
	Service	no
System-Level Info		
Architecture		RISC-V
SW Dependencies		qemu-fdt, firtool, arcilator, Verilator, dtc, SystemC, Python 3
Interfaces		
CLI		cosim, cosim-extract, cosim-gen-opt
API		NA
Input Artifacts	Artifact	RTL design
	Format	CIRCT core dialects (.hw.mlir)
	Artifact	Platform device tree
	Format	DTS
	Artifact	Co-simulation config
Output Artifacts	Format	JSON
	Artifact	Per-component co-simulation binary
	Format	ELF
	Artifact	QEMU device-tree config
	Format	DTS/DTB
Deployment		
Bundle		Docker